YData: Introduction to Data Science

Lecture 25: LLMs and Data Ethics

Overview

Clustering continued

A glimpse of LLM/chatbot and running it in Python

Ethics

Project timeline

Friday, April 24th

- Project is due on Gradescope
  - Add peer reviews to an Appendix of your project

Please also fill out the final project reflection on Canvas!

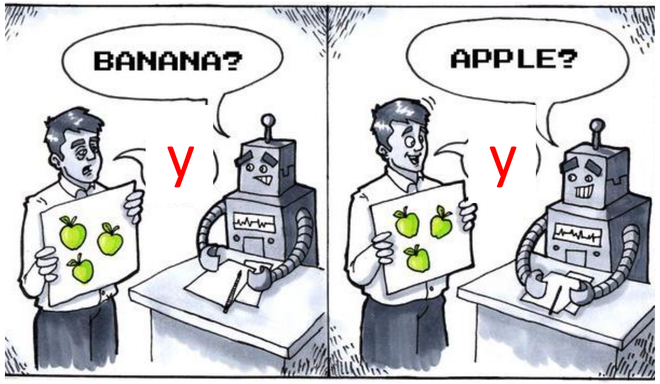
- It will be very valuable to have your feedback on how the project and class overall went

Clustering

Supervised learning and unsupervised learning

In **unsupervised learning**, we have features X, but **no** response variable y

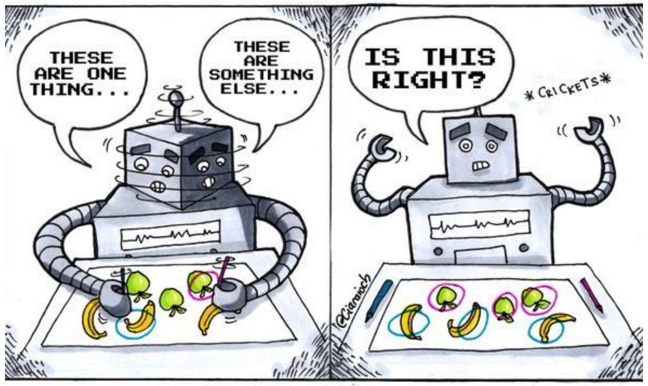
- Unsupervised learning can be useful in order to find structure in the data and to visualize patterns,
- but there is no real ground truth response variable y



Supervised Learning

1. **Clustering:** we try to group similar data points together

2. **Dimensionality reduction:** we try to find a smaller set of features that captures most of the variability in the data
   - Principal component analysis (PCA)



Unsupervised Learning

Clustering

$$n\left\{\overbrace{\begin{bmatrix} x_{11} & x_{12} & \cdots & x_{1p} \\ x_{21} & x_{22} & \cdots & x_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{np} \end{bmatrix}}^{p}\right.$$

Clustering divides n data points x_i's into subgroups

- Data points in the same group are similar/homogeneous
- Data points in different groups are different from each other

Examples:

- Examining gene expression levels to group cancer types together
- Examining consumer purchasing behavior to perform market segmentation

Clustering can be:

- **Flat**: no structure beyond dividing points into groups
- **Hierarchical**: Population is divided into smaller and smaller groups (tree like structure)

K-means clustering

K-means clustering partitions the data into ***K*** distinct, non-overlapping clusters

- i.e., each data point x_i belongs to exactly one cluster C_k

The number of clusters, ***K***, needs to be specified prior to running the algorithm

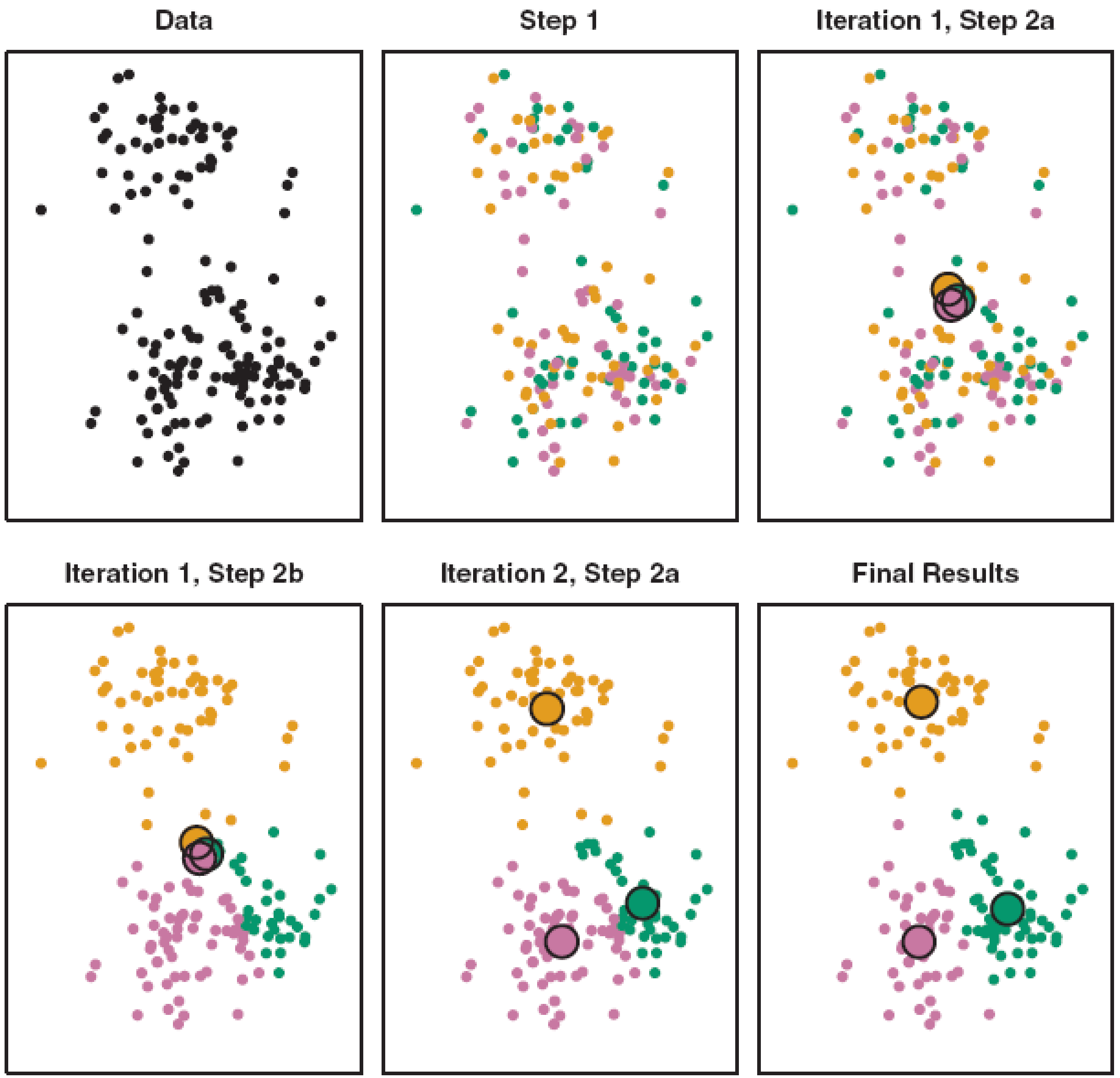
The goal is to minimize the within-cluster variation

- e.g., to make the Euclidean distance for all points within a cluster as small as possible

Finding the exact optimal solution is computationally intractable (there are k^n possible partitions), but a simple algorithm exists to find a local optimum which is often works well in practice

K-means clustering

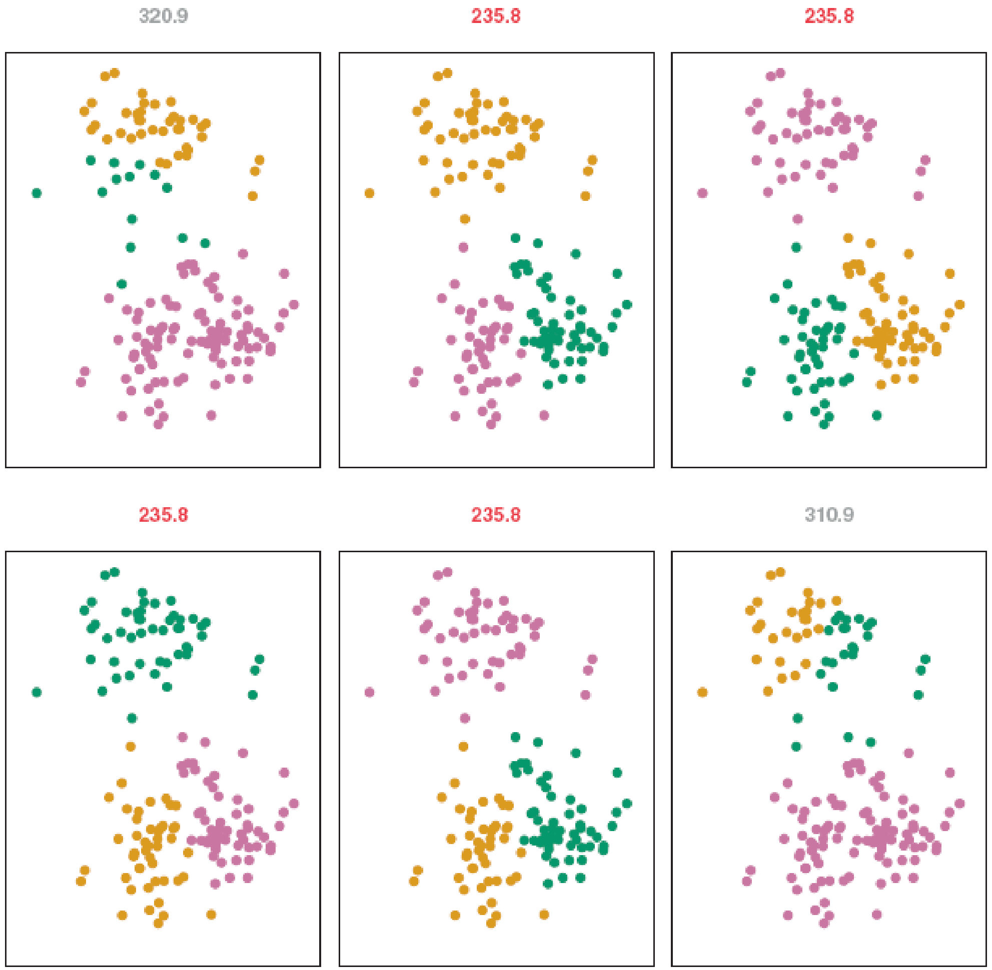
1. Randomly assign points to clusters C_k
2. Calculate cluster centers as means of points in each cluster
3. Assign points to the closest cluster center
4. Recalculate cluster center as the mean of points in each cluster
5. Repeat steps 3 and 4 until convergence



K-means clustering

Because only a local minimum is found, different random initializations will lead to different solutions

- One should run the algorithm multiple times to get better solutions



Let's continue exploring this in Jupyter!

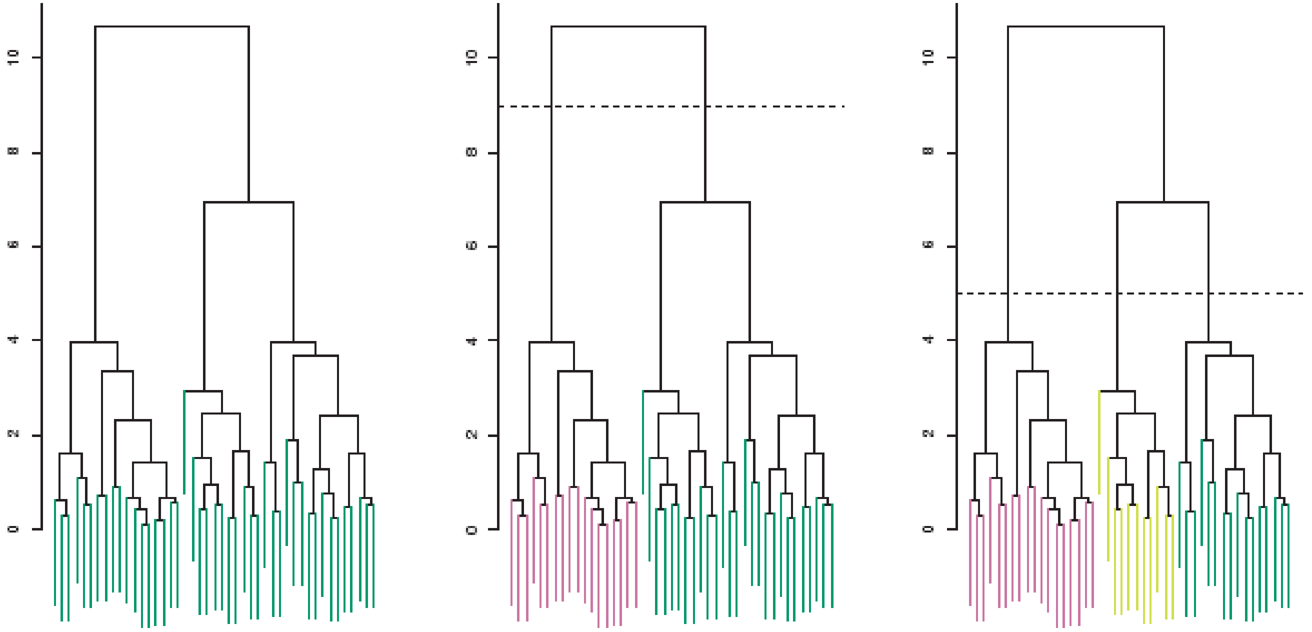
Hierarchical clustering

Hierarchical clustering

In **hierarchical clustering** we create a dendrogram which is a tree-based representation of successively larger clusters.

We can cut the dendrogram at any point to create as many clusters as desired

- i.e., don’t need to specify the number of clusters, K, beforehand



Hierarchical clustering

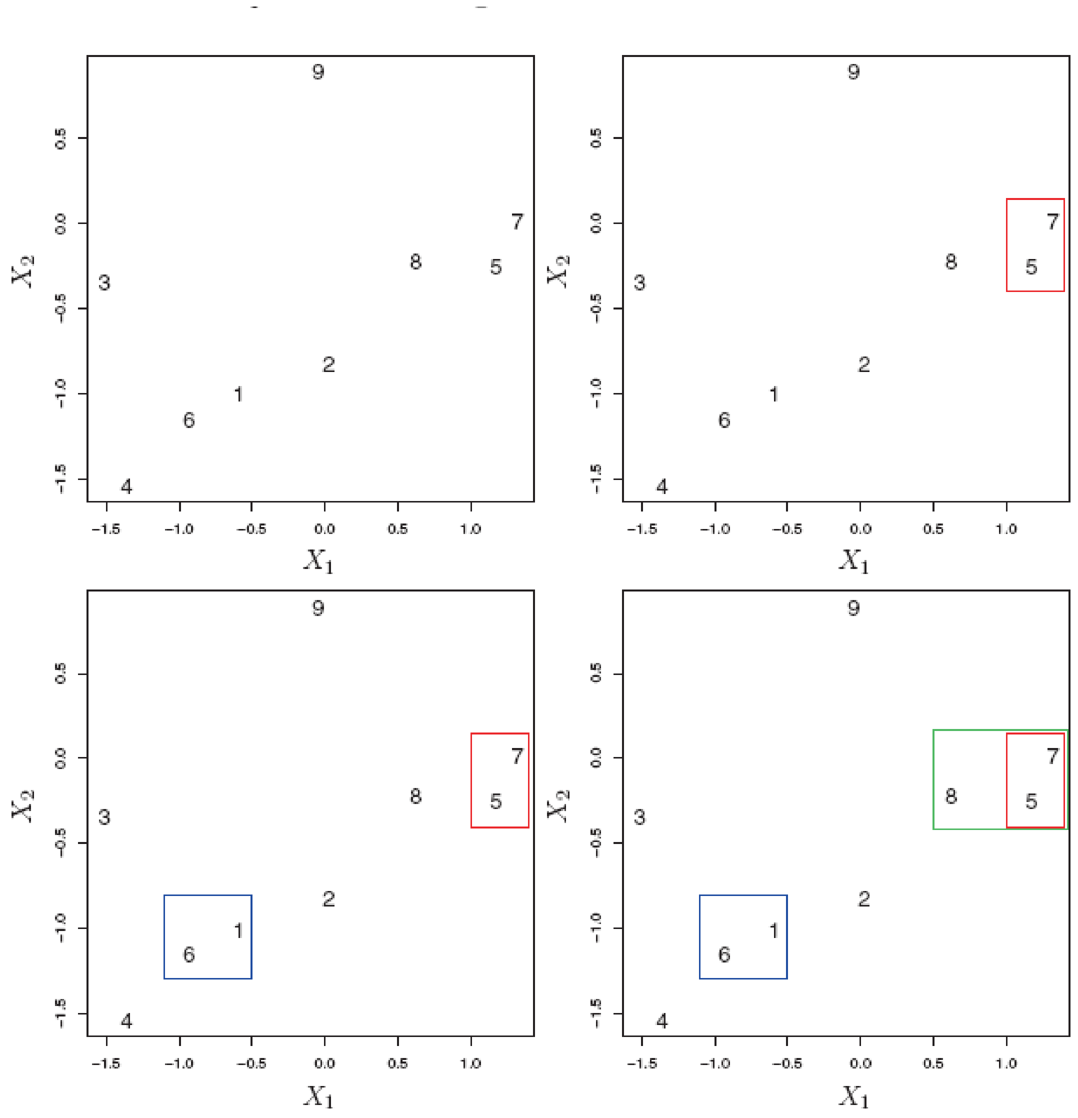
We can create a hierarchical clustering of the data using simple bottom-up agglomerative algorithm:

1. Choosing a (dis)similarity measure
   - E.g., The Euclidean distance

2. Initializing the clustering by treating each point as its own cluster

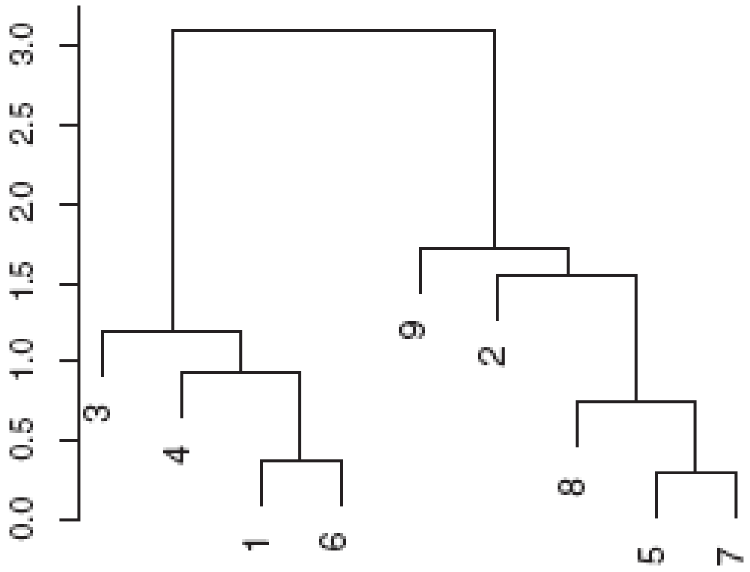
3. Successively merging the pair of clusters that are most similar
   - i.e., calculate the similarity between all pairs of clusters and merging the pair that is most similar

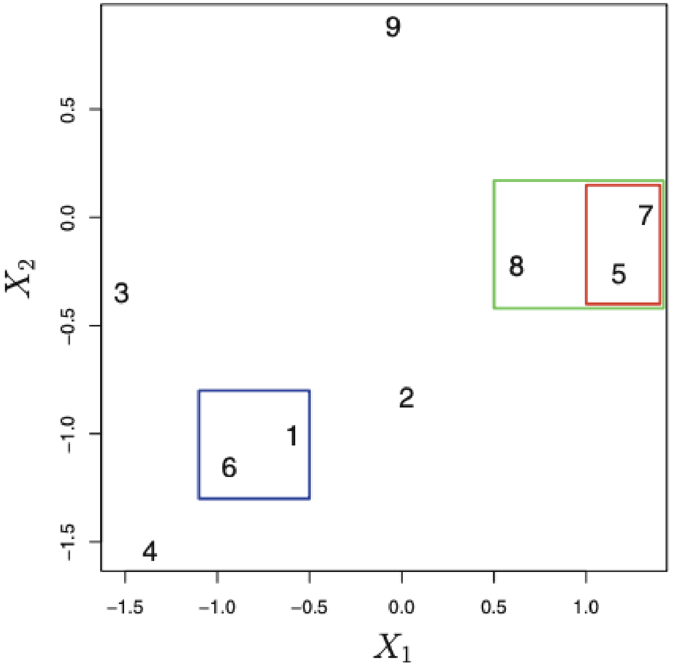
4. Stopping when all points have been merged into a single cluster



Hierarchical clustering

The vertical height that two clusters/points merge show how similar the two ***clusters*** are

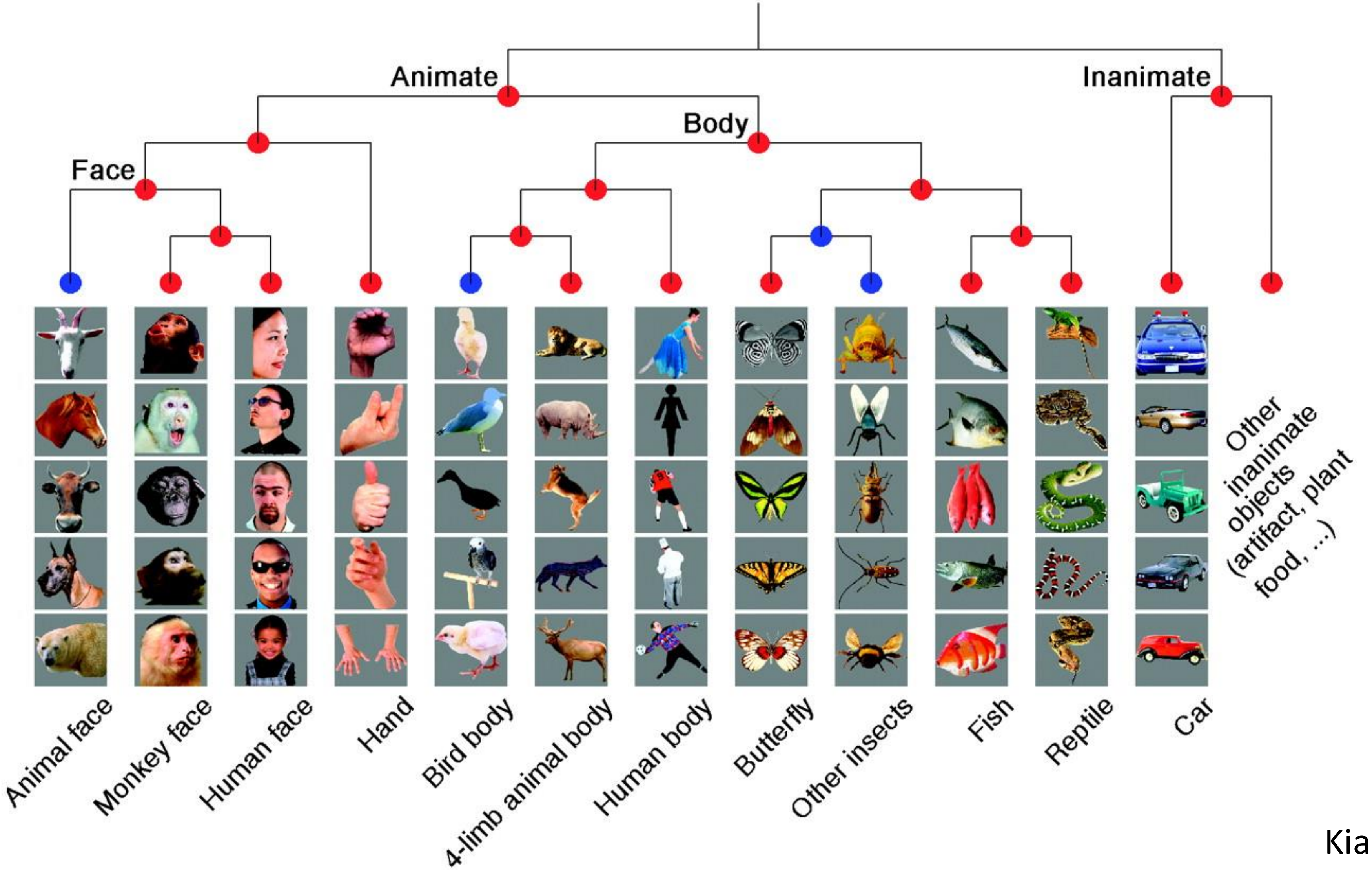




Note: horizontal distance between ***individual points*** is not important:

- point 9 is considered as similar to point 2 as it is to point 7

Hierarchical clustering example



Kiani et al, 2007

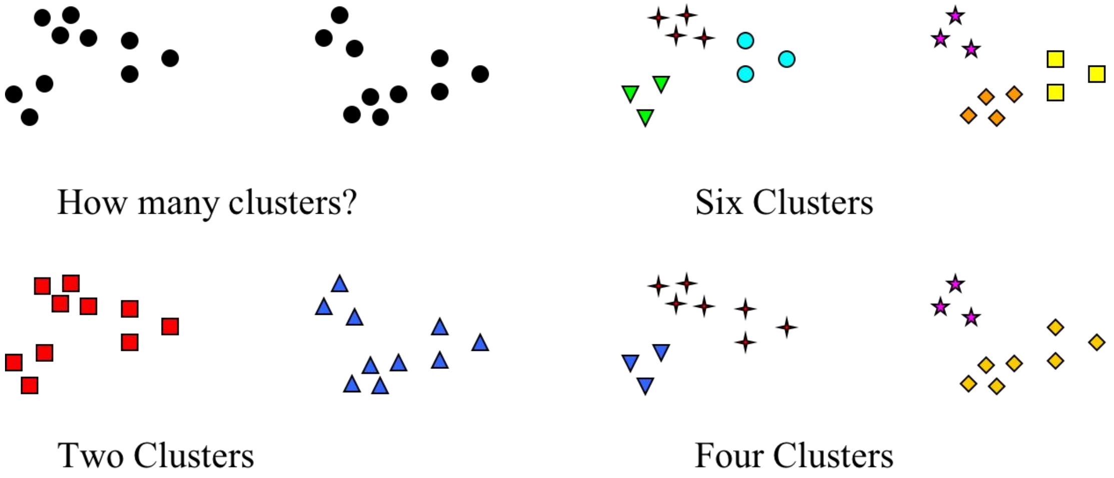
Issues with clustering

Choices made can effect the results:

- Feature normalization and/or dissimilarity measure
- K-means: choice of K
- For hierarchical cluster: linkage and cut height

Potential approaches to deal with these issues:

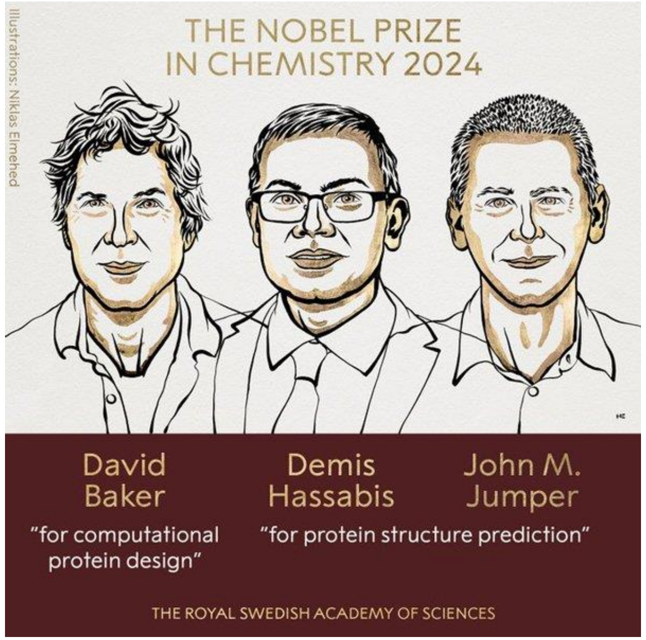
- Try a few methods and see if one gives interesting/useful results
- Validate that you get similar results on a second set of data

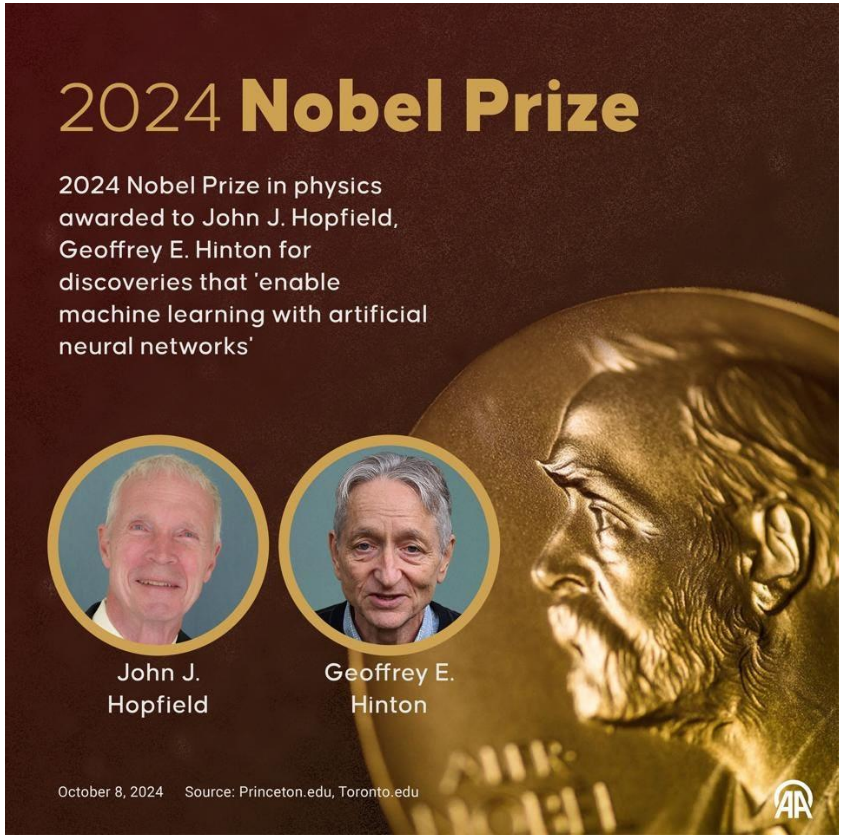


Let's explore this in Jupyter!

We did not cover…

- Many things…
- And DEEP LEARNING

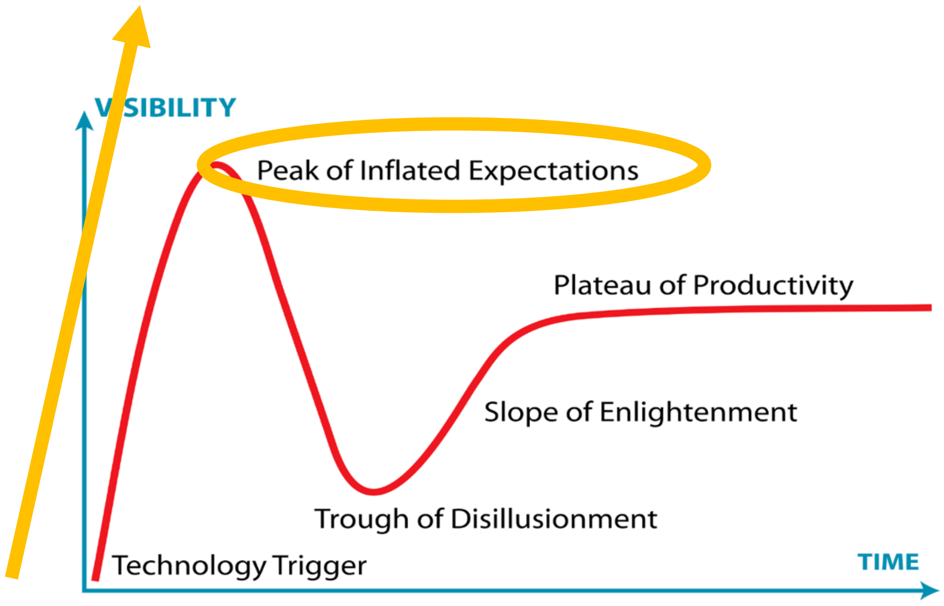




Brief discussion of Large Language Models

Large Language Models

Large language models (LLMs) are taking over the world

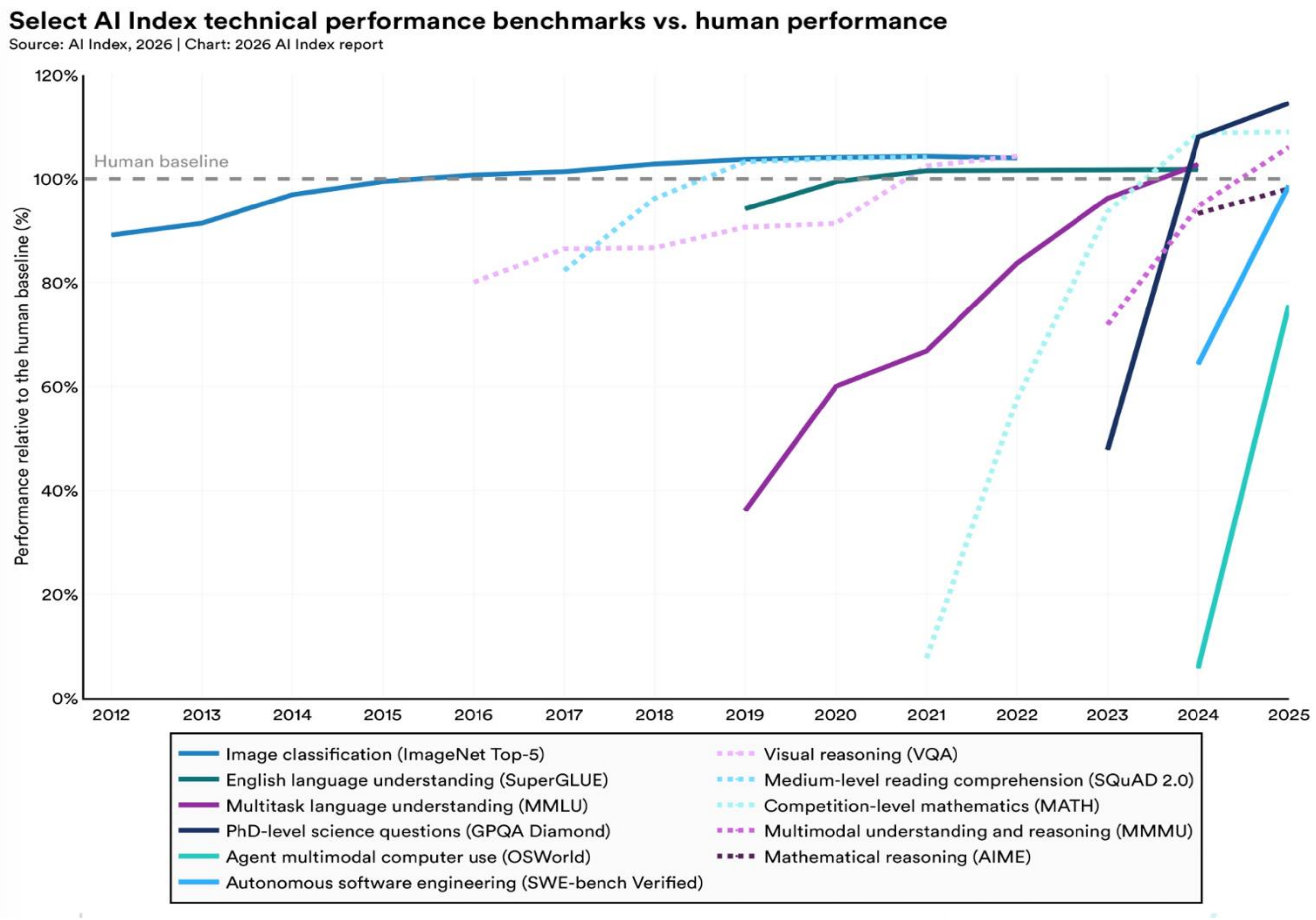


ChatGPT

Gemini

Claude

The rise of Large Language Models

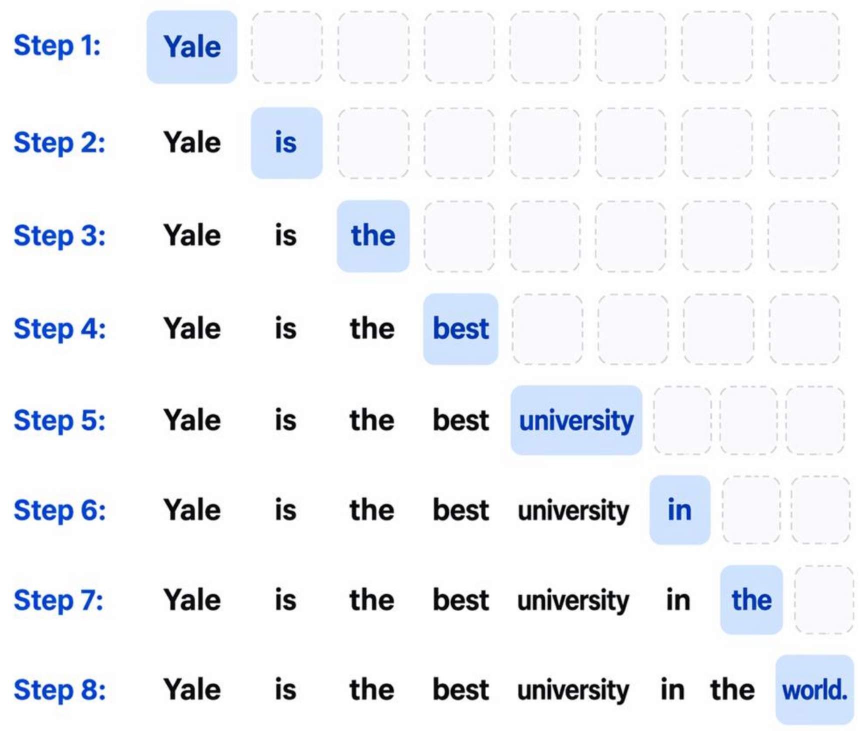


AI models are surpassing human capabilities in many domains at an accelerated rate.

https://hai.stanford.edu/ai-index/2026-ai-index-report/technical-performance

Large Language Models

- It is built on generative pre-trained transformer (GPTs).
- It functions on the mechanism of next-token prediction.
  - For intuition, think of “token” as “word”.
- It is now goes significantly beyond “language” or text, and increasingly “multi-model”: language, image, speech, video…

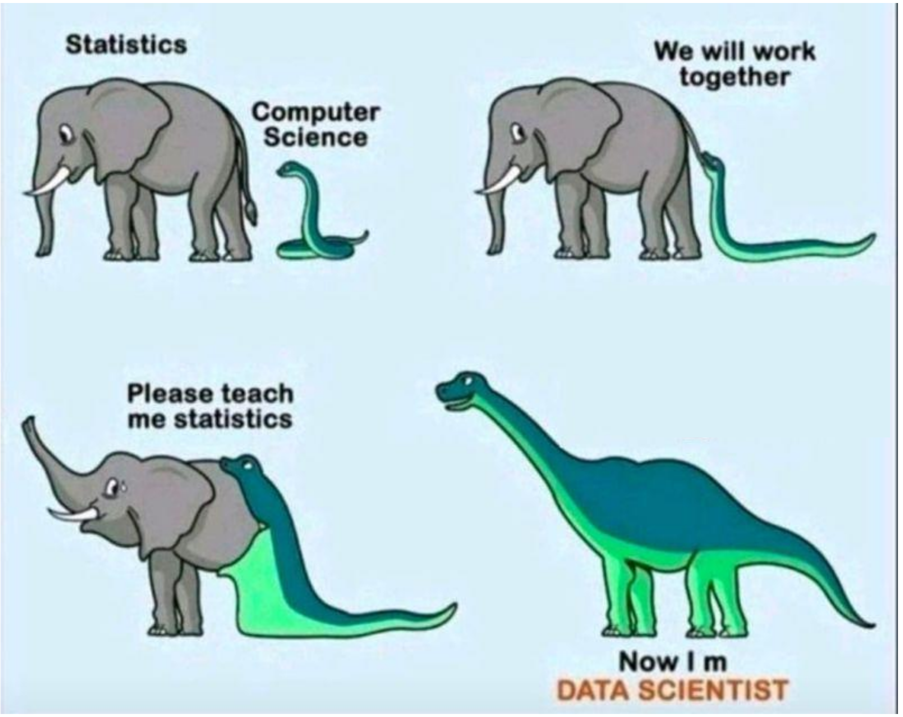


Large Language Models

LLMs can write code and [analyze data]

One can download free, open source, LLMs through the [Hugging Face platform]

Let's very briefly look at running a LLM locally on our computers…



Ethics

Ethics in Data Science

Ethics of:

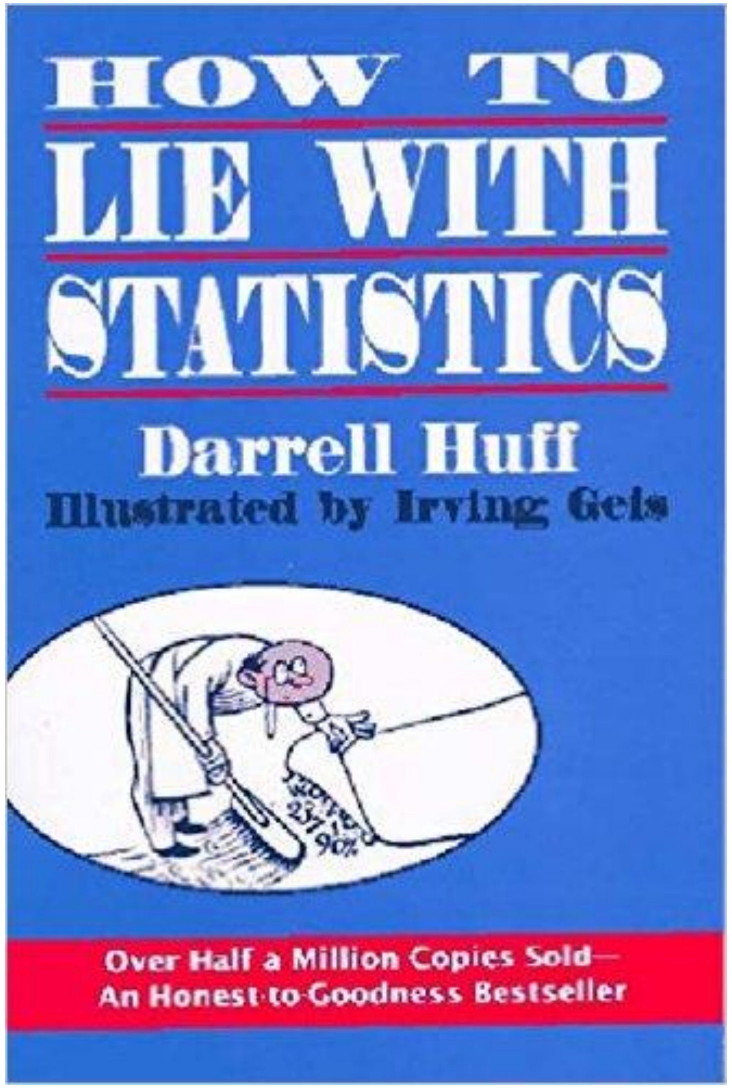
1. Data presentation
2. Using valid data
3. Data scraping TOS and privacy
4. Reproducibility
5. Ethics in Statistical analyses
6. Ethics of creating powerful tools

1. Ethics of data presentation

Data should be displayed in an honest way that gives an accurate picture of trends

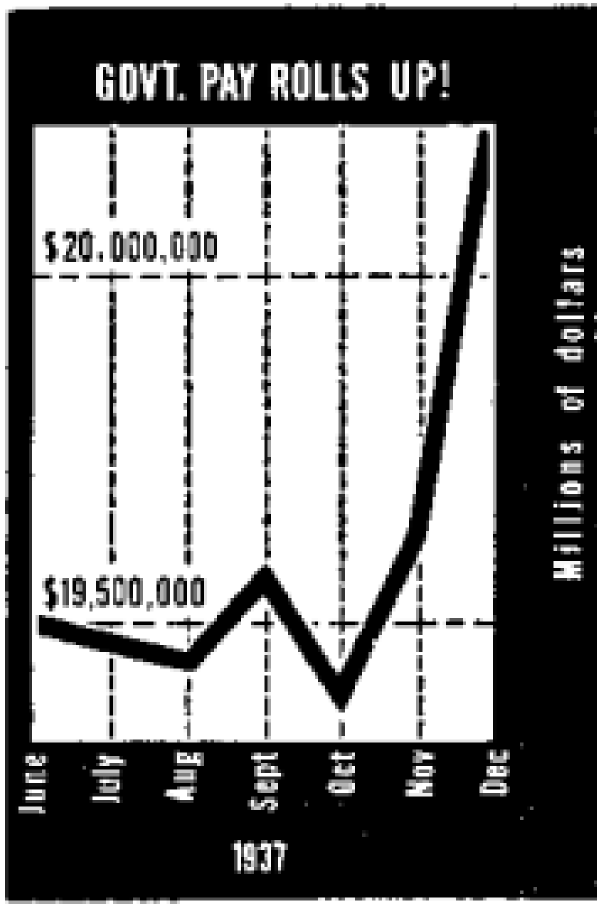
Darrell Huff wrote a classic book in the 1950’s pointing out ways that people lie with statistics

[The book was banned as training material at the VA]



Ethics of data presentation

What is potentially misleading with this figure?



From a 1938 article in Dun's Review titled 'GOVERNMENT PAY ROLLS UP!'

Case in action in 2026

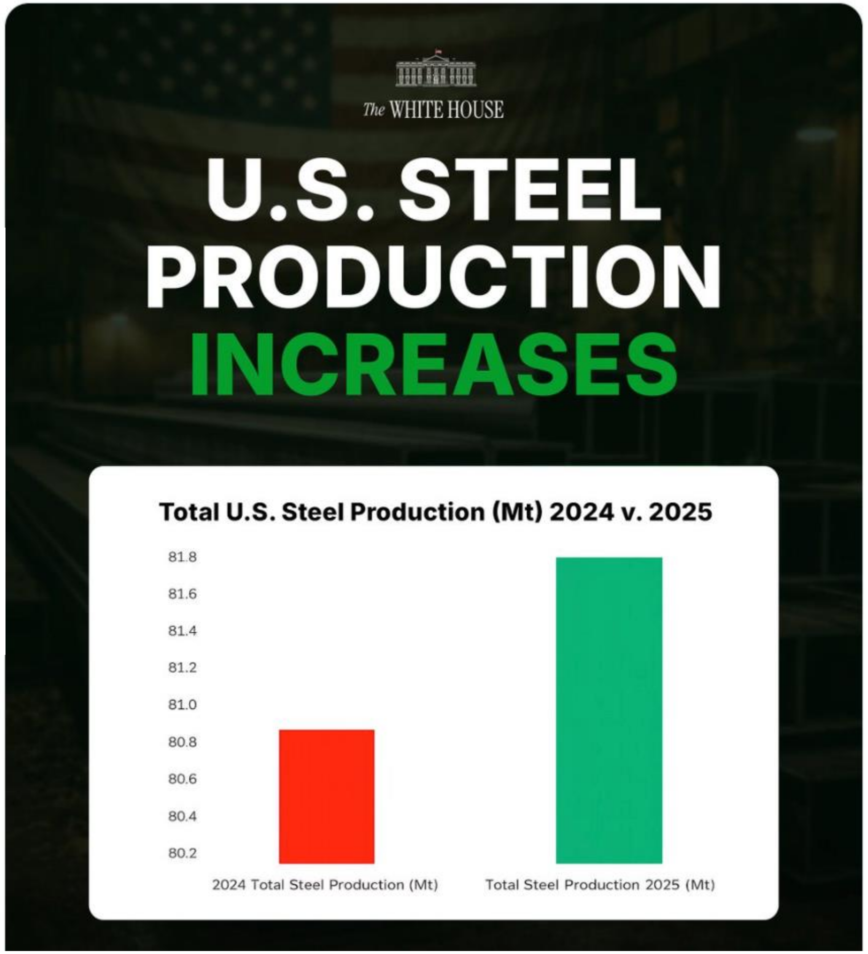
Readers added context they thought people might want to know

The vertical axis only shows values in the range 80.2Mt to 81.8Mt. This makes the 1% increase appear much larger than it actually is.

tradingeconomics.com/united-states/...

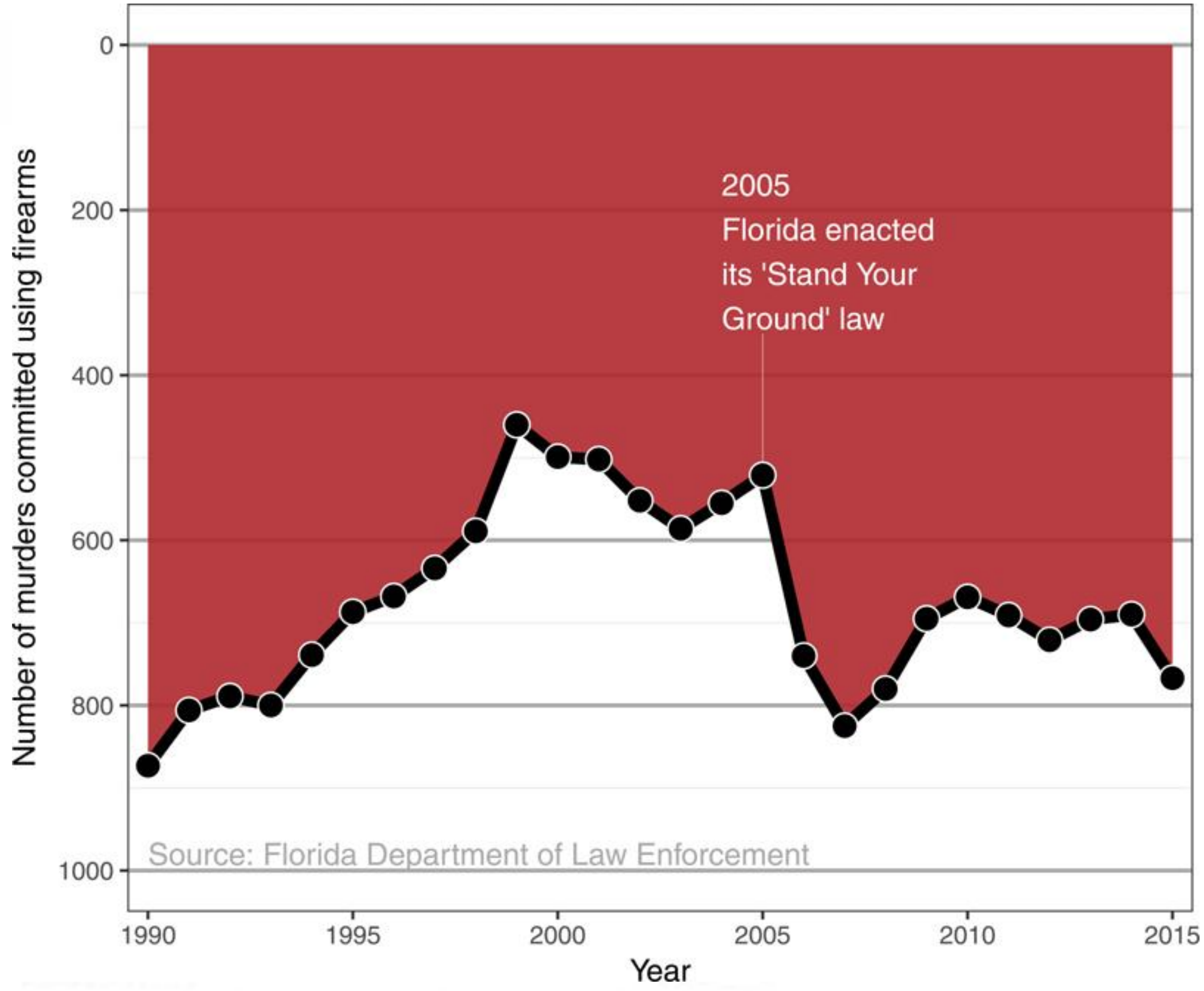
The White House @WhiteHouse

American steel is BACK. 🇺🇸



Did 'Stand Your Ground' decrease murder by firearms?

What is misleading with this figure?



2. Using valid data

Is almost everyone satisfied with Hampshire College?

Alumni Survey Results

As part of a strategic-planning process, in spring 2013 Hampshire College launched a survey of alums. Via email, the College invited 8,160 alums to fill out an online questionnaire administered by the campus's Alumni and Family Relations and Institutional Research offices. A total of 1,920 surveys were completed, yielding a response rate of 24%.

Note: The percentages in the data (below) are based on the number of responses received for each question.

To what extent do you agree with the following statements?

Strongly Agree or Agree

Hampshire encouraged me to think and work independently 99%

Hampshire encouraged me to come up with innovative ideas and solutions 95%

Hampshire improved my ability to synthesize information from across disciplines 96%

Hampshire helped shape me into a life-long learner 95%

Please rate your student experience at Hampshire.

95% Very positive or positive

3. Data scraping, terms of service and privacy

Scraping publicly available data is fine (e.g., Wikipedia) but what about scraping data if:

- It violates a website's Terms of Service?
- User privacy?

Kirkegaard and Bjerrekaer scraped okcupid and data on 68,371 users publicly available including usernames, dating preferences, etc.

- Is this ok?

Submitted: 8th of May 2016
Published: 3rd of November 2016

The OKCupid dataset: A very large public dataset of dating site users

Emil O. W. Kirkegaard* Julius D. Bjerrekær†

Open Differential Psychology

4. Reproducibility

Do scientists have an ethical obligation to make sure their research is reproducible?

nature methods

Commentary

Ethical reproducibility: towards transparent reporting in biomedical research

Reproducibility

Do scientists have an obligation to share data/code?

- What if it could hurt your career?
    - Others could prove you wrong, make new findings on your own data, etc.

What should you do if you find one of your papers is wrong?

- You need to retract the paper!

Retraction Watch

DATA COLADA

npr

NEWSLETTERS SIGN IN NPR SHOP

NEWS CULTURE MUSIC PODCASTS & SHOWS SEARCH

EDUCATION

Harvard professor who studies dishonesty is accused of falsifying data

JUNE 26, 2023 · 1:15 PM ET

Juliana Kim

Francesca Gino has been teaching at Harvard Business School for 13 years.

Maddie Meyer/Getty Images

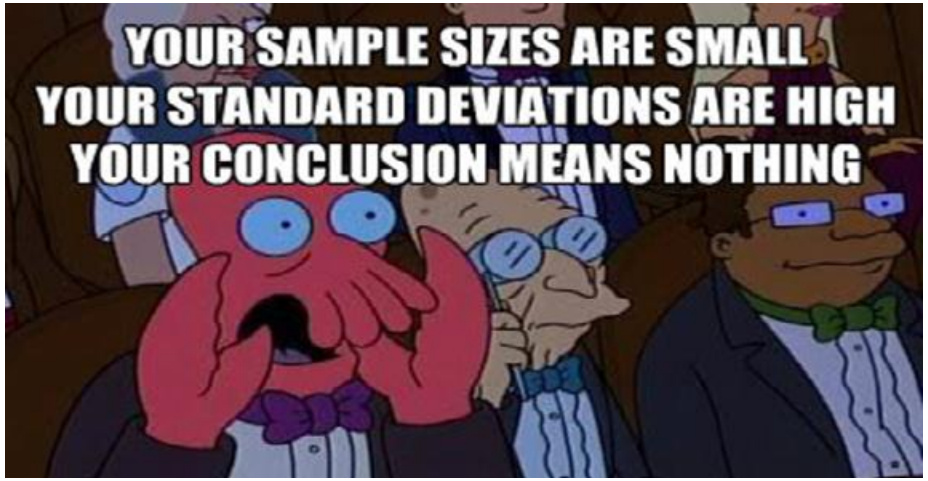
5. Ethics in Statistics

P-hacking (data dredging):

Keep trying different hypothesis tests on a data set until you reach ‘statistical significance’ ($p < 0.05$)

File drawer effect:

- Try a million studies until one is significant



DATA COLADA

6. Ethics of creating powerful tools

Some prominent people are concerned about job loss due to machine learning, or even computers posing an existential threat to humans

- Is this something we should be concerned with as Data Scientists?

Ethics in machine learning

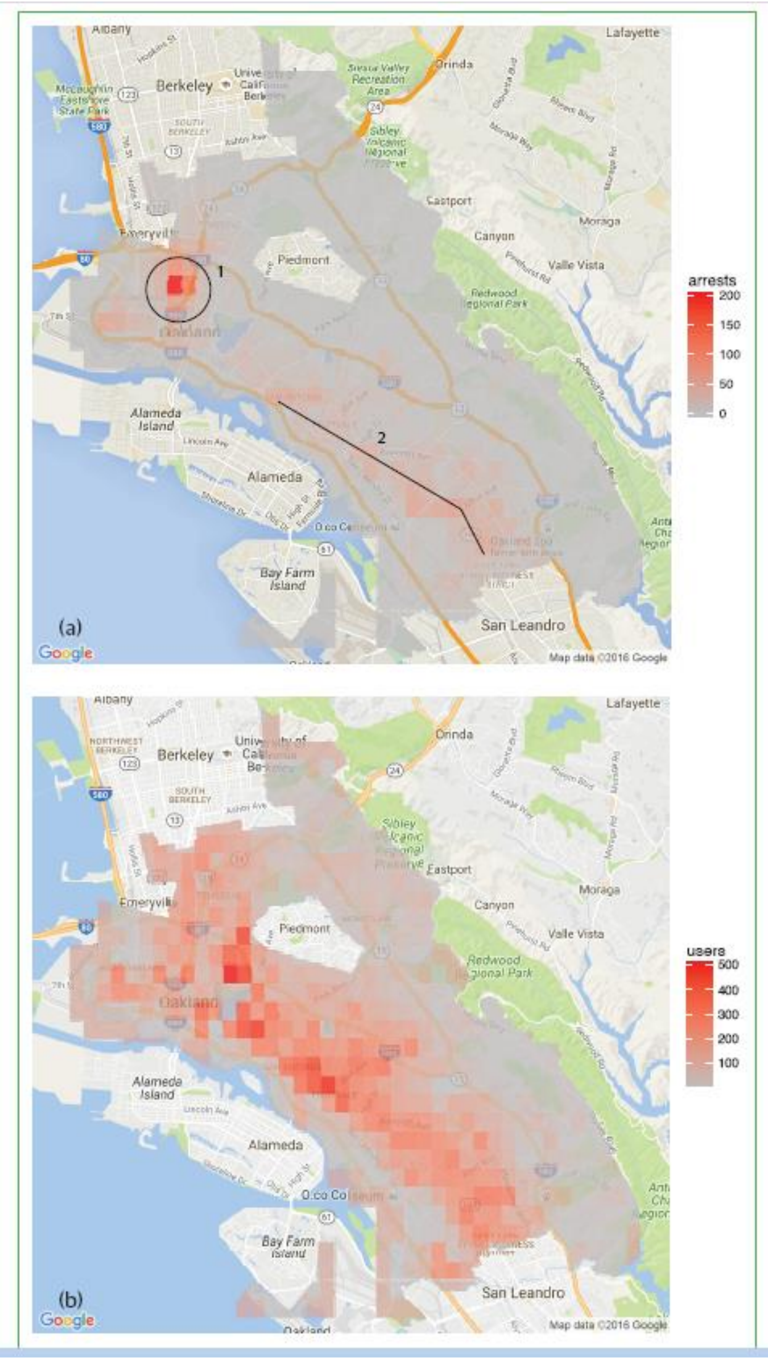
Idea: use ML to police areas with most crimes

- E.g. more police where most drug arrests have made in the past

Possible results

- Continued higher arrest rates in these areas seemingly showing the ML algorithm is working

Any potential problems with this?



Additional reading

https://www.ted.com/talks/cathy_o_neil_the_era_of_blind_faith_in_big_data_must_end